DPUDVControl

Combined distance and velocity controller for car following (e.g. for ACC assistance systems).

1. Inputs:

Activate	Activates (=1) or deactivates (=0) the controller.
TDistTarget	Desired distance to the leading vehicle [s].
vTarget	Desired velocity [m/s].
v	Current velocity [m/s].
vTO	Velocity of the leading vehicle [m/s].
dTO	Distance to the leading vehicle [m].
dMin	Minimum distance. The distanced produced of the controller will be *dMin* + *v* * *TDistTarget*.
kv	Parameter for the velocity component of the controller. The higher this value is compared to *kd* the faster the controller tries to reach *vTO*. If the leading vehicle is slower than the EGO vehicle this results in an early deceleration and a slow approach to *TDistTarget*.
kd	Parameter for the distance component of the controller. The higher this value is compared to *kv* the more the controller tries to achieve the distance *TDistTarget*. This results in later and higher decelerations.

2. Outputs:

ActivateAxControl	Activates (=1) or deactivates (=0) the acceleration controller which follows after DPUDVControl in the controller cascade (e.g. DPUAx-ControlLUT) (see image below).
axTarget	Target vehicle acceleration (m/s²). This serves as input for the acceleration controller which follows after DPUDVControl in the controller cascade (e.g. DPUAx-ControlLUT) (see image below).

3. Functional principle:

The controller is based on the publication

Paul Venhovens, Karl Naab, Bartono Adiprasito: „Stop and Go Cruise Control“, Haus der Technik e.V., Tagung H030110200, Fahrerassistenzsysteme, 29./30.11.2000

The target acceleration *u* = *axTarget* is computed by solving the following differential equation:

$$\frac{t_{dist}}{1+k_v t_{dist}}\dot{u}+u=\frac{1}{1+k_v t_{dist}}\left[k_v\Delta v+k_v k_d\Delta d+\dot{v}_{prec}\right]-a_z$$

Thereby,

tdist = TDistTarget

Δv = vTO – v

Δd = dTO

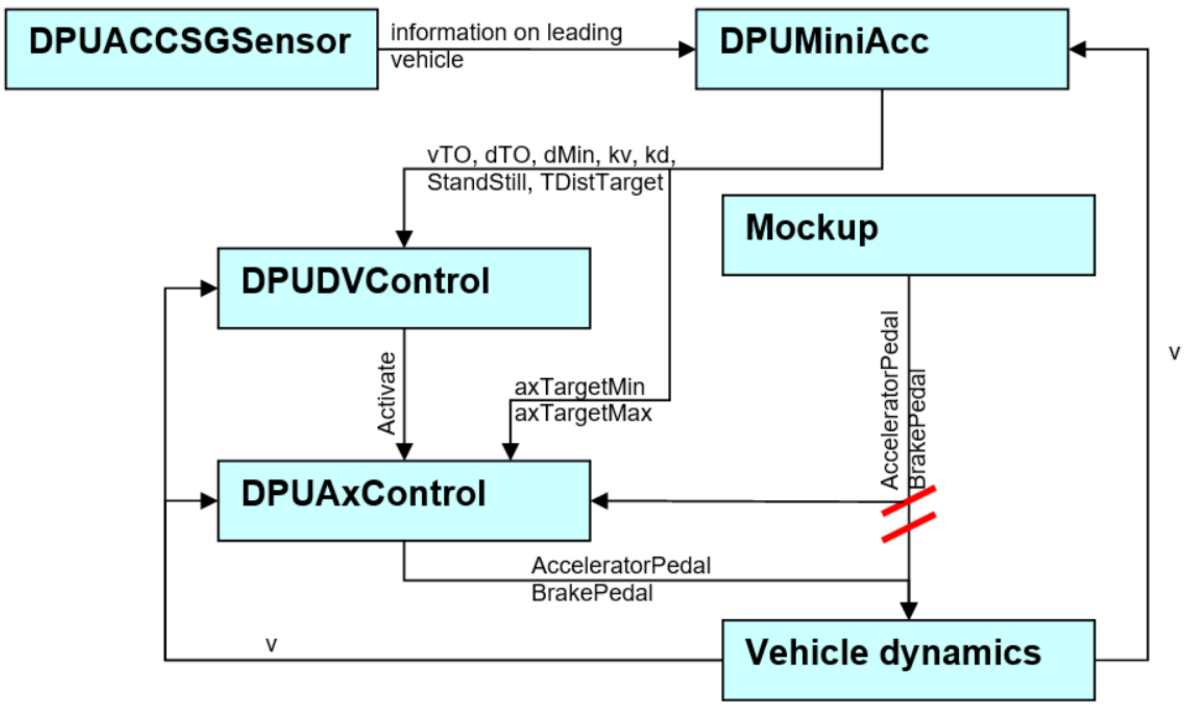
vprec = vTO

The value a_z is not used.

4. Typical controller cascade for an ACC system:

DPUDVControl is usually used as part of an ACC (Adaptive Cruise Control) system. The following image depicts the configuration for this:

Beispiel:



5. Example:

This DPU has to be used within the above context:

```
DPUMiniAcc Automation
{
	Computer = {ASS};
	Index = 101;
	TreeviewGroup = "Automation";
};

DPUDVControl Automation_DVControl
{
	Computer = {ASS};
	Index = 102;
	TreeviewGroup = "Automation";
};

DPUAxControlLUT Automation_AxControl
{
	Computer = {ASS};
	Index = 103;
	TreeviewGroup = "Automation";

	LUTAccel = "mstVedaAutoPar.BMW\accel_bmw5.lut";
	LUTDecel = "mstVedaAutoPar.BMW\decel_bmw5.lut";
};

Connections =
{
	# => Automation
	Mockup.AcceleratorPedal -> Automation.AcceleratorPedal,
	Mockup.BrakePedal -> Automation.BrakePedal,
	VDyn.v -> Automation.v,

	# => DVControl
	Automation.ActivateDVControl -> Automation_DVControl.Activate,
	Automation.vTarget -> Automation_DVControl.vTarget,
	Automation.vTO -> Automation_DVControl.vTO,
	Automation.dTO -> Automation_DVControl.dTO,
	Automation.dMin -> Automation_DVControl.dMin,
	Automation.kv -> Automation_DVControl.kv,
	Automation.kd -> Automation_DVControl.kd,
	Automation.StandStill -> Automation_DVControl.StandStill,
	VDyn.v -> Automation_DVControl.v,
	Automation.TDistTargetOut -> Automation_DVControl.TDistTarget,
	Automation.ForceAx -> Automation_DVControl.ForceAx,
	Automation.ax -> Automation_DVControl.axTargetIn,
```

```
	# => AxControl
	Automation_DVControl.ActivateAxControl -> Automation_AxControl.Acti-
vate,
	Automation_DVControl.axTarget -> Automation_AxControl.axTarget,
	Automation.axTargetMax -> Automation_AxControl.axTargetMax,
	Automation.axTargetMin -> Automation_AxControl.axTargetMin,
	Automation.axTargetDotMinOut -> Automation_AxControl.axTargetDotMin,
	Automation.axTargetDotMaxOut -> Automation_AxControl.axTargetDotMax,
	VDyn.v -> Automation_AxControl.v,
	Mockup.AcceleratorPedal -> Automation_AxControl.GasPedalIn,
	Mockup.BrakePedal -> Automation_AxControl.BrakeIn,

	# => Veda
	Automation_AxControl.GasPedalOut -> VDyn.AcceleratorPedal,
	Automation_AxControl.BrakeOut -> VDyn.BrakePedal
};
```